FRAGEN MASTERPRÜFUNG

Folgende Liste ist eine Sammlung von Fragen für die Masterprüfung für die Mastervorlesungen "Numerical Methods for Partial Differential Equations" und "Numerical Methods for Wave Propagation". Sie ist keinesfalls "vollständig" und darf gerne weiter ergänzt werden.

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1. ANALYSIS BASICS

Question 1. What is a harmonic function? State the maximum principle for harmonic function and use it to show uniqueness of a solution to

$$\begin{cases} -\Delta u = f, & \text{in } \Omega \text{ and} \\ u = g, & \text{on } \partial\Omega. \end{cases}$$

Also show that the solution of the Poisson problem depends continuously on the initial data.

Question 2. Write down the heat equation. What do the individual terms mean? State Gauß' divergence theorem and deduce Green's identities from it. Define the energy of the heat equation and show that it is conserved. Use this to show uniqueness. Also, define the energy for the wave equation and show uniqueness of solutions to the wave equation using the same methods.

Question 3. What are elliptic differential equations? What are hyperbolic differential equations? What are parabolic differential equations? When is such a problem well/ill posed? Give an example of a well posed and an example of an ill posed problem.

Question 4. Consider the Poisson problem

$$\begin{cases} -\Delta u = f, & \text{in } \Omega \text{ and} \\ u = g, & \text{on } \partial\Omega. \end{cases}$$

What is the variational formulation of this problem. Is it well posed? What theorems can be used to show this? (only theorem statements).

Question 5. Consider the wave equation in $\mathbb{R}^d$. Derive the variational formulation and show that it has a unique solution. Does a solution exist (without proof)?

Question 6. Show the following: Suppose that $u \in \mathcal{C}^2$ is a weak solution of

$$\begin{cases} -u'' = f, & \text{in } \Omega = (0,1) \text{ and} \\ u(0) = 0, & u'(1) = 0. \end{cases}$$

Write down the weak formulation to this problem. Then show that, given u as stated above, we have that u is also a classical solution to this problem.

Question 7. What is a Lipschitz domain? What is a Sobolev embedding? How is $H_0^1(\Omega)$ defined? State and prove the Poincaré-Friedrich's inequality.

Question 8. What is a the trace of a function? What does the trace theorem say? Give the main steps of the proof. Give a counterexample of a function $u \in L^2$ where the trace is not well defined or prove that the trace cannot exist on just L^2.

Question 9. Consider a bilinear form $a : V \times V \to \mathbb{R}$ and $\ell : V \to \mathbb{R}$ where V is a Hilbert space. When is a continuous? symmetric? coercive? When is ℓ continuous?

Question 10. State the theorem of Lax Milgram and give the main steps of the proof. Apply Lax-Milgram to problem

$$\begin{cases} -\Delta u = f, & \text{in } \Omega \text{ and} \\ u = 0, & \text{on } \partial\Omega. \end{cases}$$

What are à priori estimates?

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Question 11. What is the Fredholm alternative? What can deduce using it on the Helmholtz equation? (with proof!).

Question 12. What is Rellich compactness? Where do we use it? How does it differ from the Sobolev-embedding theorem?

2. Finite Elements and convergence of FEM

Question 13. Apply the finite Element method to the problem

$$\begin{cases} -u'' = f, & \text{in } \Omega = (0,1) \text{ and} \\ u(0) = 0, & u(1) = 0. \end{cases}$$

First explain how the finite element method works (from a birds eye view) and then go into detail deduce every step up to final system

$$\mathbf{A}\boldsymbol{u} = \boldsymbol{\ell}.$$

Does this system have a unique solution?

Question 14. Let $\Omega \subseteq \mathbb{R}^2$ be a Lipschitz domain. What is a triangulation of Ω? What is V_h? What do the words shape regular and quasi uniform mean? Why are they important?

Question 15. Explain the finite element method in 2D. What is the resulting system

$$\mathbf{A}\boldsymbol{u} = \boldsymbol{\ell}?$$

What is different to finite elements in $d = 1$? What is a reference element?

Question 16. Delineate how one proves convergence of the finite element method in $d = 1, 2$ with $\mathcal{P}^1$-elements and with respect to the H^1-norm. Give the main steps of the proof. What does the triangulation need to satisfy for the convergence of the FEM? What is elliptic regularity? Does elliptic regularity also holds when choosing mixed boundary conditions?

Question 17. State and prove Céa's lemma. Also, prove that the solution u_h is optimal with respect to the norm $\|v\|_{\mathrm{E}} := \sqrt{a(v,v)}$.

Question 18. Let $\hat{K} \subseteq \mathbb{R}^2$ be a fixed triangle. Given $z_1, z_2, z_3 \in \overline{\hat{K}}$ such that they are *not* co-linear and mutually distinct. We define ,

$$I_{\hat{K}} : H^2\big(\overline{\hat{K}}\big) \to \mathcal{P}^1\big(\overline{\hat{K}}\big)$$

by sending u to the polynomial that interpolates the continuous version of u at z_i, $i = 1, \dots, 3$. Show that

$$\big\|u - I_{\hat{K}} u\big\|_{H^2} \lesssim |u|_{H^2}$$

Hint: You may use Rellich's compactness theorem. If you use it, also state it.

Question 19. State and prove the scaling lemma? How is it used to derive error estimates?

Question 20. Let $\{\mathcal{T}_h\}_{h\in\mathcal{H}}$ be a triangulation. When is it shape-regular? When is it quasi-uniform? Given $K \in \mathcal{T}_h$, what are the numbers h_K, ϱ_K? How is h defined? Why do we need those terms? Explain in detail why we need shape regularity.

Question 21. How do we show convergence of higher order FEM? Note that, in order to have higher order finite elements, we need more points in the reference triangle. How do we have to choose these points. Given such a configuration of points on the reference element, prove that the interpolation of a function is unique.

How are these points chosen on a quadrilateral elements?

Question 22. Consider the Helmholtz equation in $\mathbb{R}^2$. Derive the FEM for this equation. What does the theorem of Schatz say? State and prove it. What is the dispersion/pollution error? Give an example for this behaviour. What can one do to prevent it.

3. L^2 and L^∞-error estimates

Here we estimate the error of the finite element method in L^2 and L^∞.

Question 23. Suppose that $u \in H^{r+1}$ is the weak solution of our Poisson problem. Show that

$$\|u - u_h\|_{L^2} \lesssim h^{r+1} |u|_{H^{r+1}},$$

where u_h is the numerical solution using $\mathcal{P}^r$-elements.
Hint: Use duality. To prove this result you may use that

$$\|u - I_h u\|_{H^m(\Omega)} \lesssim h^{r+1-m} |u|_{H^{r+1}(\Omega)} \quad \text{for } m = 0, 1.$$

Use this to prove

$$\|u - u_h\|_{H^1(\Omega)} \lesssim h^r |u|_{H^{r+1}(\Omega)}$$

and then use this in the proof of the question.

Question 24. Let $r = 1$ that is we consider $\mathcal{P}^1$ elements. Show that

$$\|u - u_h\|_{L^\infty} \lesssim h |u|_{H^2}$$

What prerequisites are required to show these? (Regularity of the function u, what does the mesh $\{\mathcal{T}_h\}_{h\in\mathcal{H}}$ have to satisfy?).
Hint: First show that $\|u - I_h u\|_{L^\infty(\Omega)} \lesssim h |u|_{H^2(\Omega)}$ using the Sobolev embedding theorem and show and use that on V_h the L^∞ and L^2-norm are equivalent (in some sense).

Question 25. How can we estimate the eigenvalues of the stiffness matrix $\mathbf{A}$?
Hint: Inverse inequality and show the equivalence of $|\boldsymbol{u}_h|$ and $\|u_h\|_{L^2(\Omega)}$. For the eigenvalues use Rayleigh-quotient.

How are the eigenvalues of the stiffness matrix related to the convergence/solvability of the system? What does this have to do with the eigenvalues/eigenvectors of the Laplacian $-\Delta$?

4. A posteriori error estimation and adaptivity

Question 26. Why do we need *A posteriori error estimation and adaptivity*? What is the weak residual? Define the Clément interpolant $I_h^{\text{Clement}} v$ for $v \in H^1$ and recall the notation $\mathcal{E}(K)$, $\mathcal{N}(K)$ etc. from the notes [notes, page 64].

The Clément interpolant satisfies the following result that we did not prove in the lecture: Suppose that $\{\mathcal{T}_h\}_{h\in\mathcal{H}}$ is κ-shape regular. Then one has that

$$\left\| v - I_h^{\text{Clement}} v \right\|_{L^2(K)} \leq c_1 h_K \|\nabla v\|_{L^2(\tilde{\omega}_K)}, \quad \text{and}$$
$$\left\| v - I_h^{\text{Clement}} v \right\|_{L^2(e)} \leq c_2 \sqrt{h_e} \|\nabla v\|_{L^2(\tilde{\omega}_e)}$$

where h_e is the length of the edge e and the constants c_1 and c_2 only depend on the shape regularity of the triangulation $\{\mathcal{T}_h\}_{h\in\mathcal{H}}$.

Question 27. Define the error indicators η_K and explain how they are used and why it is important that they are computable. What does the quantity η_K stand for?

Using the above result about the Clément interpolant show that

$$\|\nabla(u - u_h)\|_{L^2(\Omega)} \lesssim_\kappa \sqrt{\sum_{K\in\mathcal{T}_h} \eta_K^2}$$

Question 28. What are the two most popular marking strategies and how do they work? Write down the adaptive refinement algorithm. What do you do if $\|f\|_{L^2}$ is too complicated to compute?

Note that the previous questions gave an upper bound on the error. We also want a lower bound to have optimal control.

We again work in $\mathbb{R}^2$ and define the following two cut-off functions: the "bubble" and "edge" cut-off functions which we first define on the reference element $\hat{K}$ and its lower edge $\hat{e} = (0,1) \times \{0\}$:

$$\begin{aligned}
\psi_{\hat{K}}(\boldsymbol{\xi}) &= 27\xi_1(1-\xi_1-\xi_2), \quad \operatorname{supp}\{\psi_{\hat{K}}\} \subseteq \overline{\hat{K}} \\
\psi_{\hat{e}}(\boldsymbol{\xi}) &= 4\xi_1(1-|\xi_2|)(1-\xi_1-|\xi_2|) \quad \operatorname{supp}\{\psi_{\hat{K}}\} \subseteq \overline{\hat{K}} \cup \overline{\hat{K}_-}
\end{aligned}$$

where $\hat{K}_-$ is the mirror image of $\hat{K}$ below the x-axis. Extend this map via the pull-back map (Explain how!) and define the constant extension operator

$$E : L^2(e) \to L^2(\omega_e)$$

by setting $Ev(x) := v(x')$ if $x \in \omega_e$ is connected to $x' \in e$ via a line segment parallel to either of the two other sides of a triangle $K \in \omega_e$.

Question 29. First prove that for every $K \in \mathcal{T}$ and $e \subseteq \partial K$ we have

$$\begin{aligned}
\|v\|_{L^2(K)} &\le c_1 \|\sqrt{\psi}\, v\|_{L^2(K)}, && \text{for all } v \in \mathcal{P}^1(K) \\
\|\nabla(\psi_K v)\|_{L^2(K)} &\le c_2 h_K^{-1} \|v\|_{L^2(K)}, && \text{for all } v \in \mathcal{P}^1(K) \\
\|v\|_{L^2(e)} &\le c_3 \|\sqrt{\psi_e} v\|_{L^2(e)}, && \text{for all } v \in \mathcal{P}^1(e) \\
\|\nabla(\psi_e E v)\|_{L^2(\omega_e)} &\le c_4\, h_e^{-1/2} \|v\|_{L^2(e)}, && \text{for all } v \in \mathcal{P}^1(e) \\
\|\psi_e E v\|_{L^2(\omega_e)} &\le c_5 \sqrt{h_e} \|v\|_{L^2(e)}, && \text{for all } v \in \mathcal{P}^1(e),
\end{aligned}$$

then derive a lower bound for the error estimators with a constant only depending on the shape regularity of the mesh.

Question 30. Prove that the refinement algorithm converges with Dörfler Marking converges. In particular prove the estimate

$$\|\nabla(u-u_{\ell+1})\|_{L^2}^2 \le \left(1 - \frac{\vartheta}{c_\star^2 (c^\star)^2}\right) \|\nabla(u-u_\ell)\|_{L^2}^2$$

under the assumption that f is constant on any triangle of the initial triangulation $\mathcal{T}^{(0)}$. Note that the constants $c^\star, c_\star \ge 1$ come from the upper and lower bound of the error estimators and depend only on the shape regularity of the initial triangulation $\mathcal{T}^{(0)}$.

5. Discontinuous Galerkin FEM

Question 31. What are Discontinuous Galerkin (DG) Finite Element Methods? Explain why they are important. Derive the weak formulation for DGFEM. What do you have to add to the bilinear form to make it symmetric and coercive? (no proof required yet). Write down this modified bilinear form a_{DG}.

Question 32. How is the mesh dependent DG energy norm $\|\cdot\|_h$ defined. Prove that a is coercive if $\sigma_{\min} < \min_e \sigma_e$ large enough. Deduce that the discrete DG formulation has a unique solution in this case.
Hint: Discrete trace inequalities

Question 33. Show that that for shape regular triangulations we have that (here $r = 1$),

$$\|u - u_h\|_h \lesssim h \|u\|_{H^2}, \quad h \le h_0,$$

that is, the DG method convergences.
Hint: Previous question and continuous trace inequalities. Deduce, using consistency, that Galerkin orthogonality must hold for a_{DG}. Then estimate the subtracted terms in the bilinear form from above using Cauchy-Schwartz and Young's inequality.

6. FEM on time dependent problems

Question 34. Derive the weak formulation of the heat equation. Discretize first in space and obtain the semi-discrete solution of the heat equation. How do we discretize this further? What methods can you use to discretize in time? What convergence rate do you obtain
Hint: The best one is Crank Nicolson.

Question 35. What is the Ritz-projection Π_h? How it can be used to solve for e.g., the heat equation? Show that the Ritz-projection is linear, continuous and commutates with time-derivatives.

Question 36. Discretize the heat equation in space. Prove that the following estimate holds:

$$\|u(t) - u_h(t)\|_{L^2} \le \|(\mathrm{id} - \Pi_h)u(t)\|_{L^2} + \|u_{0,h} - \Pi_h u_0\|_{L^2} + \int_0^t \big\|(\mathrm{id} - \Pi_h)u'(s)\big\|_{L^2} \,\mathrm{d}s,$$

where Π_h denotes the Ritz-projection and $u_{0,h}$ is some approximation of the initial datum u_0, e.g., L^2-projection or some sort of interpolant. Use this estimate to show that given the right conditions we have that

$$\|u(t) - u_h(t)\|_{L^2} \le C(T+1)h^2 \|u\|_{C_t H_x^2}$$

What are these conditions?

Question 37. Derive the semi-discrete FEM method for the wave equation. Show that the discrete energy is conserved. Discretize the semi-discrete equation via Leapfrog. Define the energy for the fully discrete method and show its conservation. Why do we need that? Do we have an analogue to a CFL condition?

Question 38. Consider the semi-discrete FEM method for the wave equation. What is the problem with this formulation? What is mass lumping? Explain mass lumping for $d = 1, 2$. For $d = 2$ consider triangular and quadrilateral elements. What is Gauss-Lobato quadrature?

Question 39. State and show convergence of the semi-discrete FEM for the wave equation.
Hint: What do you have to do with the initial data? For the proof derive an analogue to Galerkin orthogonality and use to get an expression for

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}\left\{ \big\|e'(t)\big\|_{L^2}^2 + a\big(e(t), e(t)\big) \right\}$$

involving $\rho = (u - I_h u)(t)$. Then integrate over the above expression and estimate.

7. Waves

Question 40. What is a wave? Given the Euler equations (conservation of mass and momentum),

$$\frac{\partial}{\partial t}\rho + \operatorname{div}(\rho \boldsymbol{v}) = 0, \quad \text{and} \quad \rho\frac{\partial \boldsymbol{v}}{\partial t} + \rho(\boldsymbol{v}\cdot\nabla)\boldsymbol{v} = -\nabla p + \boldsymbol{F}$$

linearize the Euler equation about the state $\rho \equiv \rho_0$, $\boldsymbol{v} \equiv \boldsymbol{0}$ and derive the wave equation from this.

Question 41. Consider the wave equation in $\mathbb{R}^d = \mathbb{R}$, that is $d = 1$. Derive D'Alamberts solution formula. What is the domain of dependence. How can we use this formula when dealing with just $\mathbb{R}_+$ or $\mathbb{R}_-$. What do the initial conditions have to satisfy for the extension technique to work?

Question 42. What are time-harmonic waves? Why are they important? What are planar or spherical waves and why are they called that? Give a mathematical intuition.

Starting from a time continuous wave derive the Helmholtz equation and its weak formulation. Why are we not able to apply Lax-Milgram. What can we use instead?

Question 43. What is a DtN map? Where is it used? Give an example of an application (with proof that the resulting problem has the same solution as the old one).

Question 44. What is a time-harmonic scattering problem? What Sommerfeld's radiation condition? Give some intuition behind Sommerfeld (1D example is enough).

Question 45. Derive the DtN map for a ball $B_R(0)$. Show that given the DtN map the problems

$$\begin{cases} \Delta u + k^2 u = 0, & \text{in } \Omega, \\ u = g, & \text{on } \Gamma = \partial D \\ \frac{\partial u}{\partial r} = M[u], & \text{on } \partial B_R(0). \end{cases} \qquad \begin{cases} \Delta u + k^2 u = 0, & \text{in } \Omega_\infty := \mathbb{R}^2 \setminus \overline{D}, \\ u = g, & \text{on } \Gamma = \partial D \\ \lim_{r \to +\infty} \sqrt{r}(\partial_r u - \mathrm{i}kv) = 0, & \end{cases}$$

(where $D \subseteq \mathbb{R}^2$ is a bounded obstacle) agree in Ω.
Hint: For the proof of the last statement you may use that a solution of the Helmholtz equation is analytic.

Question 46. Explain the DtN finite difference method for the Helmholtz equation and write down the resulting linear system.
Hint: Work in polar coordinates.

Question 47. Explain the DtN FEM for the Helmholtz equation and write down the resulting linear system. What is the problem with the DtN method?
Hint: Work in polar coordinates and see the next question.

Explain why we need the modified DtN condition. Prove that the problem with the modified DtN method has at most one solution.

Question 48. Consider the Helmholtz equation

$$\begin{cases} \Delta u + k^2 u = 0 & \text{in } \Omega \\ u = 0 & \text{on } \Gamma \\ \frac{\partial u}{\partial r} = M^N[u] & \text{on } B = \partial B_R(0), \end{cases} \tag{1}$$

where

$$M^N[u] = \frac{1}{\pi} \sum_{n=0}^{N}{}' \alpha_n \int_0^{2\pi} u(R, \varphi) \cos\big(n(\theta - \varphi)\big)\, \mathrm{d}\varphi.$$

The sum with the prime stands for a sum where the zeroth coefficient is multiplied by 1/2. Take $\Gamma = \partial B_\rho(0)$ with $\rho \in (0, R)$ and $g(\theta) = \cos(m\theta)$. Show that in general this system does not have a unique solution. What do we do against this?

Question 49. List some advanteges and disadvantages of the DtN boundary condition. Derive *The local boundary conditions of Bayliss-Gunzburger - Turkel (BGT)* and explain the advantages of that method. Show that

$$\|u - v\|_{L^2(B)} \lesssim \frac{1}{kR^2}$$

where u is the solution to the problem with Sommerfeld condition and v the solution to problem with $B_1[v] = 0$ on $\partial B_R(0)$.

8. Finite Differences for Waves

Question 50. Derive the Leap-Frog method for the wave equation with c variable. Derive the CFL-condition for stability. Show energy conservation of that method and use it to show convergence. First show that the energy is bounded by first three terms plus some additional terms. Then show convergence.

Question 51. What is "von Neumann" stability. How can we use it to show for example the convergence of the Lax-Friedrich's method. Consider a general explicit one-step scheme of the form

$$v_j^{m+1} = \sum_{\ell=\alpha}^{\beta} Q_\ell v_{j+\ell}^m, \quad m \geq 0$$

with Q only depend on $r = c\Delta t / \Delta x$. Explain the concepts of stability and accuracy and why they are important for the convergence.

Question 52. State and show the Lax-Equivalence theorem for a numerical scheme of the form

$$v_j^{m+1} = \sum_{\ell=\alpha}^{\beta} Q_\ell v_{j+\ell}^m, \quad m \geq 0.$$

What do the numbers Q_ℓ have to satisfy for stability? (with proof!). Prove that the Lax-Friedrich's scheme is stable. Use this to show that the Lax-Friederich's method converges (when the CFL-condition is met).

Question 53. What is the Lax-Wndroff-method? Why is it interesting? Show that it is stable and use "von Neumann" stability analysis to deduce that it converges. What is the order of convergence (again given the right CFL-condition).

Question 54. Consider a problem of the form

$$\begin{aligned} \partial_t u + c(x)\partial_x u &= 0, && \text{in } (0,T)\times\mathbb{R} \\ u(t=0,x) &= f(x), && \text{for all } x \in \mathbb{R}. \end{aligned} \tag{2}$$

What is an upwinding method? Why is it interesting (i.e., what would be a better method on first glance)? Show stability and convergence.

Given a wave equation in 2D how can we transform it into a problem of the form (2)?

9. Nonreflecting Boundary Conditions for the Wave Equation

Question 55. Consider the equation

$$\begin{cases} \partial_t \boldsymbol{u} + \mathbf{A}\partial_x \boldsymbol{u} = \mathbf{0}, & \text{in } \mathbb{R}\times(0,1) \\ \boldsymbol{u}(t=0,x) = \boldsymbol{f}(x), & \text{for all } x\in\mathbb{R}. \end{cases}$$

where $\boldsymbol{u}(t,x) \in \mathbb{R}^m$ and $\mathbf{A} \in \mathbb{R}^{m\times m}$. Suppose that $\mathbf{A}$ is diagonalizable. Decouple the equations using diagonalization. Derive non-reflecting boundary conditions and show that resulting system can be transformed into the previous one an vice versa.

Question 56. Consider the hyperbolic Cauchy problem

$$\begin{cases} \partial_t \boldsymbol{u} + \mathbf{A}\partial_x \boldsymbol{u} = \mathbf{0}, & \text{for } (t,x)\in\mathbb{R}_+\times\mathbb{R} \\ \boldsymbol{u}(t=0,x) = \boldsymbol{f}(x), & \text{for all } x\in\mathbb{R}. \end{cases} \tag{3}$$

with $\mathbf{A} \in \mathbb{R}^{m\times m}$ with $\mathbf{A} = \mathbf{S\Lambda S}^{-1}$ with

$$\mathbf{\Lambda} = \begin{bmatrix} \mathbf{\Lambda}^{\mathrm{I}} & & \\ & \mathbf{\Lambda}^{\mathrm{II}} & \\ & & \mathbf{\Lambda}^{\mathrm{III}} \end{bmatrix}$$

real diagonal with $\mathbf{\Lambda}^{\mathrm{I}} > \mathbf{0} \in \mathbb{R}^{r\times r}$, $\mathbf{\Lambda}^{\mathrm{II}} < \mathbf{0} \in \mathbb{R}^{s\times s}$ and $\mathbf{\Lambda}^{\mathrm{III}} = \mathbf{0}$. Suppose that $\operatorname{supp}\{f\} \subseteq \Omega = (0,1)$ and

$$\mathbf{L}_0 := \left[\; \mathbf{I}_{r\times r} \;\middle|\; \mathbf{0} \;\middle|\; \mathbf{0} \;\right] \mathbf{S}^{-1} \in \mathbb{R}^{r\times m}, \quad \mathbf{L}_1 := \left[\; \mathbf{I}_{r\times r} \;\middle|\; \mathbf{0} \;\middle|\; \mathbf{0} \;\right] \mathbf{S}^{-1} \in \mathbb{R}^{s\times m}$$

Show that then the solution to the initial-boundary value problem

$$\begin{cases} \frac{\partial}{\partial t}\boldsymbol{u} + \mathbf{A}\frac{\partial}{\partial x}\boldsymbol{u} = 0, & t>0,\ 0<x<1, \\ \boldsymbol{u}(0,x) = \boldsymbol{f}(x), & \\ \mathbf{L}_0\boldsymbol{u}(t,0) = \mathbf{0}, & t>0, \\ \mathbf{L}_1\boldsymbol{u}(t,1) = \mathbf{0}, & t>0, \end{cases}$$

coincides with the restriction to $(0,1)$ of the solution to the Cauchy problem (3).

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